

3	(a) (sum of/total) potential energy and kinetic energy of (all) molecules/particles reference to random (distribution)	M1 A1	[2]
	(b) (i) no heat enters (gas)/leaves (gas)/no heating (of gas)	B1	
	work done by gas (against atmosphere as it expands)	M1	
	internal energy decreases	A1	[3]
	(ii) volume decreases so work done on ice/water (allow work done negligible because ΔV small)	B1	
	heating of ice (to break rigid forces/bonds)	M1	
	internal energy increases	A1	[3]